Supplementary Figure and Tables

Label-efficient choice of proxy tracks for variant-effect prediction

All tables are generated by `paper/make_supplement.py` from the result tables under `results/a1` in the code repository. Spearman correlations are between predicted and measured effects of held-out variants; differences are paired within a unit (Kircher genomic region, or Siraj locus with its two cell types averaged). Siraj-derived tables are released under CC-BY-NC 4.0. Table S6 has two parts (S6a, S6b). Table S15 is copied from the result files named in its source.

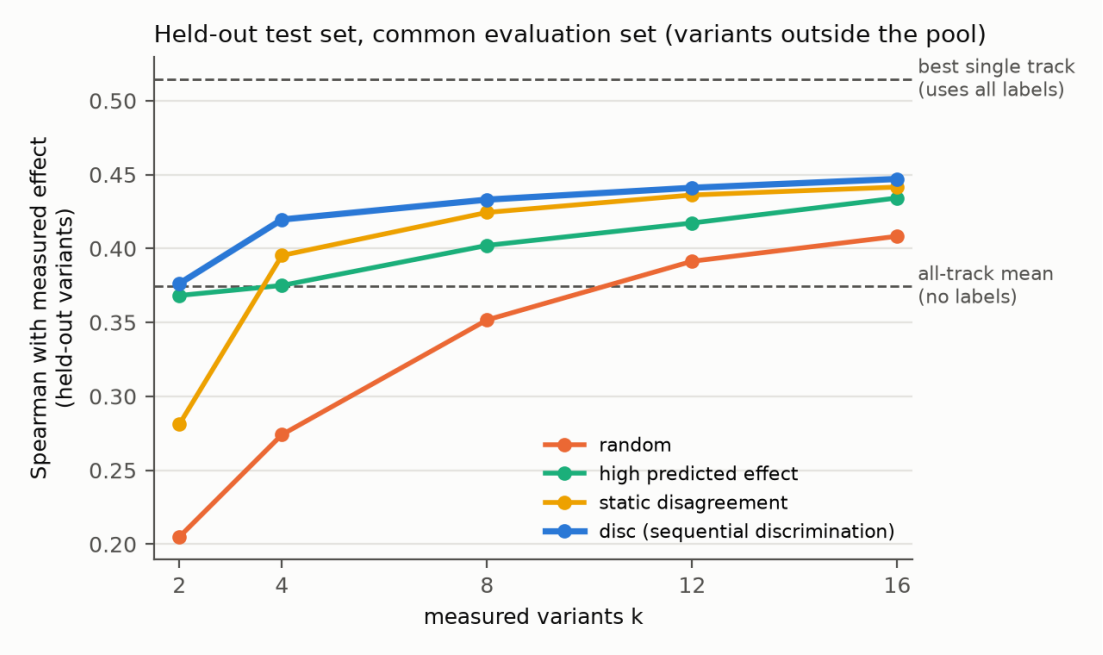


Figure S1: Kircher test set, unprofiled scenario: Spearman correlation on held-out variants against the number of measured variants k. All panels use the same track posterior. Dashed lines: the label-free all-track mean and the best single track chosen with all labels (a reference).

Table S1: Siraj test (125 loci): effects relative to the measured reference oligo (primary) and centred on all oligos of the element as published (original analysis). P1–P4 are the pre-registered hypotheses; all held under both definitions. Secondary rows give the mean difference over loci and its 95% bootstrap interval. S3b was added with the reference-relative labels.

	Quantity	Reference-relative	Published centring
P1 $k = 8$	mean D (lower bound)	+0.009 (+0.005)	+0.010 (+0.005)
P2 $k = 8$	harmed (upper bound)	3/125 (0.061)	3/125 (0.061)
P1 $k = 16$	mean D (lower bound)	+0.019 (+0.013)	+0.020 (+0.013)
P2 $k = 16$	harmed (upper bound)	1/125 (0.037)	1/125 (0.037)
P3 $k = 8$	v2 − all-track mean (P)	+0.047 (P=3e-12)	+0.048 (P=8e-13)
P4 $k = 16$	v2 − related mean (P)	+0.040 (P=3e-19)	+0.041 (P=1e-19)
S1 C1 k=8	v2 disc panel − v2 random panel	+0.016 [+0.012 to +0.021]	+0.018 [+0.013 to +0.022]
S1 C3 k=8	v2 disc panel − v2 random panel	+0.096 [+0.087 to +0.104]	+0.093 [+0.085 to +0.101]
S2 C1 k=8	v2 − meta_ridge on the same disc panel	+0.007 [+0.003 to +0.011]	+0.008 [+0.004 to +0.012]
S3 C1 k=8	matched_cal (random panel) − matched	−0.082 [−0.090 to −0.075]	−0.079 [−0.086 to −0.072]
S3b C1 k=8	matched_cal (disc panel) − matched	−0.010 [−0.015 to −0.006]	–
S5 C1 k=8	family prior − uniform prior	−0.002 [−0.004 to −0.000]	−0.002 [−0.003 to +0.000]
S7 HEPG2 C1 k=8	v2 − matched	+0.016 [+0.007 to +0.026]	+0.017 [+0.007 to +0.026]
S7 HEPG2 C3 k=8	v2 − all_mean	+0.050 [+0.029 to +0.074]	+0.052 [+0.031 to +0.076]
S7 K562 C1 k=8	v2 − matched	+0.003 [−0.001 to +0.007]	+0.003 [−0.002 to +0.007]
S7 K562 C3 k=8	v2 − all_mean	+0.043 [+0.029 to +0.058]	+0.044 [+0.031 to +0.059]

Table S4: Acquisition comparators (post hoc; frozen v2 inference, 50 repeats). Siraj views C1–C3; Kircher test regions with matched tracks kept (deploy) or without an anchor (no_anchor). Reading: 95% interval within ±0.02 = equivalent; above or below 0 = first or second better.

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C1	8	disc − batch_kb	125	+0.002	−0.000 to +0.004	equivalent
Siraj	C2	8	disc − batch_kb	125	+0.005	+0.003 to +0.007	equivalent
Siraj	C3	8	disc − batch_kb	125	+0.004	+0.002 to +0.006	equivalent
Siraj	C1	8	disc − disc_static	125	+0.007	+0.004 to +0.012	equivalent
Siraj	C2	8	disc − disc_static	125	+0.014	+0.008 to +0.020	first better
Siraj	C3	8	disc − disc_static	125	+0.023	+0.017 to +0.029	first better
Siraj	C1	8	eig − disc	125	+0.000	−0.001 to +0.002	equivalent
Siraj	C2	8	eig − disc	125	+0.001	−0.000 to +0.002	equivalent
Siraj	C3	8	eig − disc	125	+0.000	−0.001 to +0.002	equivalent
Siraj	C1	8	batch_kb − disc_static	125	+0.005	+0.002 to +0.010	equivalent
Siraj	C2	8	batch_kb − disc_static	125	+0.009	+0.004 to +0.014	equivalent
Siraj	C3	8	batch_kb − disc_static	125	+0.019	+0.014 to +0.025	first better
Siraj	C1	8	disc − high_effect	125	+0.010	+0.005 to +0.016	equivalent
Siraj	C2	8	disc − high_effect	125	+0.017	+0.011 to +0.024	first better
Siraj	C3	8	disc − high_effect	125	+0.028	+0.020 to +0.035	first better
Siraj	C1	8	disc − disagree	125	+0.013	+0.008 to +0.018	equivalent
Siraj	C2	8	disc − disagree	125	+0.017	+0.011 to +0.023	first better

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C3	8	disc − disagree	125	+0.032	+0.024 to +0.041	first better
Siraj	C1	16	disc − batch_kb	125	+0.004	+0.002 to +0.007	equivalent
Siraj	C2	16	disc − batch_kb	125	+0.008	+0.006 to +0.010	equivalent
Siraj	C3	16	disc − batch_kb	125	+0.006	+0.004 to +0.009	equivalent
Siraj	C1	16	disc − disc_static	125	+0.009	+0.005 to +0.013	equivalent
Siraj	C2	16	disc − disc_static	125	+0.016	+0.013 to +0.021	first better
Siraj	C3	16	disc − disc_static	125	+0.016	+0.012 to +0.021	first better
Siraj	C1	16	eig − disc	125	+0.001	+0.000 to +0.002	equivalent
Siraj	C2	16	eig − disc	125	+0.001	+0.000 to +0.002	equivalent
Siraj	C3	16	eig − disc	125	+0.000	−0.001 to +0.001	equivalent
Siraj	C1	16	batch_kb − disc_static	125	+0.005	+0.002 to +0.009	equivalent
Siraj	C2	16	batch_kb − disc_static	125	+0.008	+0.005 to +0.012	equivalent
Siraj	C3	16	batch_kb − disc_static	125	+0.010	+0.006 to +0.014	equivalent
Siraj	C1	16	disc − high_effect	125	+0.012	+0.007 to +0.018	equivalent
Siraj	C2	16	disc − high_effect	125	+0.023	+0.018 to +0.029	first better
Siraj	C3	16	disc − high_effect	125	+0.027	+0.021 to +0.033	first better
Siraj	C1	16	disc − disagree	125	+0.011	+0.006 to +0.016	equivalent
Siraj	C2	16	disc − disagree	125	+0.015	+0.011 to +0.020	first better
Siraj	C3	16	disc − disagree	125	+0.020	+0.016 to +0.025	first better
Kircher	deploy	8	disc − batch_kb	11	+0.004	−0.001 to +0.009	equivalent
Kircher	no_anchor	8	disc − batch_kb	11	+0.003	−0.004 to +0.010	equivalent
Kircher	deploy	8	disc − disc_static	11	+0.006	+0.000 to +0.011	equivalent
Kircher	no_anchor	8	disc − disc_static	11	+0.015	−0.001 to +0.035	undetermined
Kircher	deploy	8	eig − disc	11	+0.003	+0.001 to +0.005	equivalent
Kircher	no_anchor	8	eig − disc	11	+0.002	−0.001 to +0.005	equivalent
Kircher	deploy	8	batch_kb − disc_static	11	+0.002	−0.003 to +0.006	equivalent
Kircher	no_anchor	8	batch_kb − disc_static	11	+0.012	+0.001 to +0.029	first better
Kircher	deploy	8	disc − high_effect	11	+0.004	−0.007 to +0.017	equivalent
Kircher	no_anchor	8	disc − high_effect	11	+0.029	+0.001 to +0.059	first better
Kircher	deploy	8	disc − disagree	11	+0.005	−0.009 to +0.023	undetermined
Kircher	no_anchor	8	disc − disagree	11	+0.019	−0.003 to +0.041	undetermined
Kircher	deploy	16	disc − batch_kb	11	+0.005	+0.000 to +0.012	equivalent
Kircher	no_anchor	16	disc − batch_kb	11	+0.008	+0.002 to +0.013	equivalent
Kircher	deploy	16	disc − disc_static	11	+0.020	+0.000 to +0.048	first better
Kircher	no_anchor	16	disc − disc_static	11	+0.013	−0.002 to +0.028	undetermined
Kircher	deploy	16	eig − disc	11	+0.004	+0.001 to +0.007	equivalent
Kircher	no_anchor	16	eig − disc	11	+0.002	−0.001 to +0.006	equivalent
Kircher	deploy	16	batch_kb − disc_static	11	+0.015	−0.003 to +0.043	undetermined
Kircher	no_anchor	16	batch_kb − disc_static	11	+0.005	−0.011 to +0.018	equivalent
Kircher	deploy	16	disc − high_effect	11	+0.022	−0.004 to +0.055	undetermined
Kircher	no_anchor	16	disc − high_effect	11	+0.001	−0.021 to +0.023	undetermined
Kircher	deploy	16	disc − disagree	11	+0.020	+0.002 to +0.041	first better
Kircher	no_anchor	16	disc − disagree	11	+0.008	−0.004 to +0.019	equivalent

Table S2: Directional accuracy (post hoc): fraction of significant held-out variants ($P < 10^{-5}$, |effect| ≥ 0.1) whose predicted sign is correct. Labelled methods use the posterior-average prediction; *no offset* drops the fitted offsets (diagnostic). Interval: 95% bootstrap over units; last column, units better/worse.

Set	Contrast	First	Second	n	Acc. 1	Acc. 2	Diff.	95% CI	+/−
Siraj	P1	disc@8	matched@0	125	0.874	0.867	+0.007	−0.010 to +0.024	63/60
Siraj	P1 (S4)	disc@16	matched@0	125	0.886	0.867	+0.019	+0.001 to +0.037	66/57
Siraj	P3	disc@8	all_mean@0	125	0.864	0.809	+0.055	+0.032 to +0.078	76/48
Siraj	P4	disc@16	related@0	125	0.873	0.834	+0.039	+0.020 to +0.057	80/45
Siraj	S1	disc@8	random@8	125	0.864	0.789	+0.074	+0.061 to +0.089	107/18
Siraj	S3	random+matched_cal@8	matched@0	125	0.817	0.867	−0.050	−0.068 to −0.032	30/95
Siraj	S3b	disc+matched_cal@8	matched@0	125	0.865	0.867	−0.002	−0.021 to +0.017	57/66
Kircher (v1)	H1	disc@8	random@8	11	0.675	0.694	−0.018	−0.105 to +0.042	7/4
Kircher (v1)	H2	disc@8	all_mean@0	11	0.675	0.774	−0.099	−0.267 to +0.046	4/7
Kircher (v2)	v2 C1 k=8	disc@8	matched@0	11	0.706	0.839	−0.133	−0.264 to −0.026	2/9
Kircher (v2)	v2 C1 k=16	disc@16	matched@0	11	0.730	0.839	−0.108	−0.230 to −0.010	3/8
Kircher (v2, no offset)	v2 C1 k=8	disc@8	matched@0	11	0.768	0.839	−0.071	−0.207 to +0.011	5/4
Kircher (v2, no offset)	v2 C1 k=16	disc@16	matched@0	11	0.790	0.839	−0.048	−0.158 to +0.027	6/3
Kircher (v2, no offset)	v2 C3 k=8	disc@8	all_mean@0	11	0.743	0.774	−0.031	−0.188 to +0.107	8/3
Kircher (v2, no offset)	v2 C3 k=8 vs random	disc@8	random@8	11	0.743	0.700	+0.042	−0.029 to +0.101	9/2
Kircher (v2)	v2 C3 k=8	disc@8	all_mean@0	11	0.694	0.774	−0.079	−0.232 to +0.055	4/7
Kircher (v2)	v2 C3 k=8 vs random	disc@8	random@8	11	0.694	0.693	+0.002	−0.066 to +0.052	8/3

Table S3: Siraj test repeated with, for each locus, the AlphaGenome checkpoint that held out its genomic fold (labels relative to the reference oligo). Columns give the pre-registered contrast with the distilled checkpoint (ALL_FOLDS) and with the held-out checkpoint, and the paired change with its 95% bootstrap interval over loci.

Hypothesis	ALL_FOLDS	Held-out fold	Change [95% CI]	Verdict
P1 $k = 8$	+0.009	+0.024	+0.014 [+0.008 to +0.022]	holds
P1 $k = 16$	+0.019	+0.037	+0.018 [+0.009 to +0.027]	holds
P3 $k = 8$	+0.047	+0.050	+0.003 [−0.006 to +0.013]	holds
P4 $k = 16$	+0.040	+0.053	+0.013 [+0.006 to +0.021]	holds
P2 $k = 8$	3/125 harmed	1/125 harmed	–	holds
P2 $k = 16$	1/125 harmed	1/125 harmed	–	holds

Table S5: CODA- and MODEL SELECTOR-style baselines (post hoc; 50 repeats). Both use the calibrated likelihood, no metadata and the information-gain panel; CODA uses a consensus prior ($\kappa = 10$, fixed on development elements), MODEL SELECTOR a uniform prior ($\kappa = 0$). *Single track*: the most probable track, the methods' own output; *average*: their posterior average. Positive: units where the first is better.

Data	View	k	Contrast	n	Mean	95% CI	+	Reading
Siraj	C1	8	v2 − coda (single track)	125	+0.075	+0.062 to +0.089	111	first better
Siraj	C1	8	v2 − model_selector (single track)	125	+0.074	+0.064 to +0.084	118	first better
Siraj	C1	8	coda: average − single track	125	+0.016	+0.012 to +0.021	99	first better
Siraj	C1	8	v2 − coda (average)	125	+0.059	+0.045 to +0.072	105	first better

Data	View	k	Contrast	n	Mean	95% CI	+	Reading
Siraj	C1	16	v2 − coda (single track)	125	+0.045	+0.038 to +0.054	111	first better
Siraj	C1	16	v2 − model_selector (single track)	125	+0.042	+0.035 to +0.049	112	first better
Siraj	C1	16	coda: average − single track	125	+0.009	+0.005 to +0.012	92	equivalent
Siraj	C1	16	v2 − coda (average)	125	+0.037	+0.028 to +0.045	107	first better
Siraj	C2	8	v2 − coda (single track)	125	+0.029	+0.019 to +0.039	88	first better
Siraj	C2	8	v2 − model_selector (single track)	125	+0.036	+0.029 to +0.045	99	first better
Siraj	C2	8	coda: average − single track	125	+0.018	+0.014 to +0.022	103	first better
Siraj	C2	8	v2 − coda (average)	125	+0.011	+0.002 to +0.021	70	first better
Siraj	C2	16	v2 − coda (single track)	125	+0.017	+0.010 to +0.023	88	first better
Siraj	C2	16	v2 − model_selector (single track)	125	+0.018	+0.013 to +0.023	90	first better
Siraj	C2	16	coda: average − single track	125	+0.012	+0.009 to +0.015	98	equivalent
Siraj	C2	16	v2 − coda (average)	125	+0.005	−0.001 to +0.011	70	equivalent
Siraj	C3	8	v2 − coda (single track)	125	+0.034	+0.027 to +0.042	106	first better
Siraj	C3	8	v2 − model_selector (single track)	125	+0.044	+0.039 to +0.049	121	first better
Siraj	C3	8	coda: average − single track	125	+0.017	+0.013 to +0.021	105	first better
Siraj	C3	8	v2 − coda (average)	125	+0.018	+0.011 to +0.024	83	first better
Siraj	C3	16	v2 − coda (single track)	125	+0.028	+0.023 to +0.034	114	first better
Siraj	C3	16	v2 − model_selector (single track)	125	+0.030	+0.026 to +0.034	115	first better
Siraj	C3	16	coda: average − single track	125	+0.012	+0.009 to +0.015	98	equivalent
Siraj	C3	16	v2 − coda (average)	125	+0.016	+0.012 to +0.021	93	first better
Kircher	C1	8	v2 − coda (single track)	11	+0.082	+0.006 to +0.149	8	first better
Kircher	C1	8	v2 − model_selector (single track)	11	+0.070	−0.003 to +0.138	8	undetermined
Kircher	C1	8	coda: average − single track	11	+0.026	+0.015 to +0.037	9	first better
Kircher	C1	8	v2 − coda (average)	11	+0.056	−0.019 to +0.120	8	undetermined
Kircher	C1	16	v2 − coda (single track)	11	+0.056	+0.013 to +0.099	9	first better
Kircher	C1	16	v2 − model_selector (single track)	11	+0.061	+0.013 to +0.108	7	first better
Kircher	C1	16	coda: average − single track	11	+0.017	+0.005 to +0.029	7	first better
Kircher	C1	16	v2 − coda (average)	11	+0.039	−0.007 to +0.081	8	undetermined
Kircher	C3	8	v2 − coda (single track)	11	+0.051	+0.018 to +0.087	8	first better
Kircher	C3	8	v2 − model_selector (single track)	11	+0.042	+0.005 to +0.075	8	first better
Kircher	C3	8	coda: average − single track	11	+0.025	+0.015 to +0.034	9	first better
Kircher	C3	8	v2 − coda (average)	11	+0.026	−0.005 to +0.059	7	undetermined
Kircher	C3	16	v2 − coda (single track)	11	+0.030	+0.015 to +0.045	9	first better
Kircher	C3	16	v2 − model_selector (single track)	11	+0.028	+0.002 to +0.051	8	first better
Kircher	C3	16	coda: average − single track	11	+0.019	+0.007 to +0.032	8	first better
Kircher	C3	16	v2 − coda (average)	11	+0.011	−0.008 to +0.028	6	undetermined

Table S6b: EIG convergence check, accuracy: differences in Spearman correlation of the posterior-average prediction (same pools and 50 repeats; 95% bootstrap interval over units).

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C1	8	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C2	8	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C3	8	E2 − E0	125	+0.0004	−0.001 to +0.001	equivalent
Siraj	C1	8	E2 − disc	125	−0.0000	−0.001 to +0.001	equivalent
Siraj	C2	8	E2 − disc	125	+0.0006	−0.001 to +0.002	equivalent
Siraj	C3	8	E2 − disc	125	+0.0007	−0.001 to +0.002	equivalent
Siraj	C1	8	E0 − disc	125	+0.0002	−0.001 to +0.002	equivalent
Siraj	C2	8	E0 − disc	125	+0.0008	−0.000 to +0.002	equivalent
Siraj	C3	8	E0 − disc	125	+0.0003	−0.001 to +0.002	equivalent
Siraj	C1	8	E1 − E0	125	−0.0001	−0.001 to +0.001	equivalent
Siraj	C2	8	E1 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C3	8	E1 − E0	125	+0.0003	−0.001 to +0.001	equivalent
Siraj	C1	16	E2 − E0	125	−0.0005	−0.001 to +0.000	equivalent
Siraj	C2	16	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C3	16	E2 − E0	125	−0.0001	−0.001 to +0.001	equivalent
Siraj	C1	16	E2 − disc	125	+0.0007	+0.000 to +0.001	equivalent
Siraj	C2	16	E2 − disc	125	+0.0009	+0.000 to +0.002	equivalent
Siraj	C3	16	E2 − disc	125	+0.0001	−0.000 to +0.001	equivalent
Siraj	C1	16	E0 − disc	125	+0.0012	+0.000 to +0.002	equivalent
Siraj	C2	16	E0 − disc	125	+0.0011	+0.000 to +0.002	equivalent
Siraj	C3	16	E0 − disc	125	+0.0001	−0.001 to +0.001	equivalent
Siraj	C1	16	E1 − E0	125	−0.0005	−0.001 to +0.000	equivalent
Siraj	C2	16	E1 − E0	125	−0.0002	−0.001 to +0.000	equivalent
Siraj	C3	16	E1 − E0	125	+0.0000	−0.001 to +0.001	equivalent
Kircher	deploy	8	E2 − E0	11	+0.0007	−0.001 to +0.002	equivalent
Kircher	no_anchor	8	E2 − E0	11	−0.0001	−0.002 to +0.002	equivalent
Kircher	deploy	8	E2 − disc	11	+0.0038	+0.002 to +0.006	equivalent
Kircher	no_anchor	8	E2 − disc	11	+0.0022	−0.000 to +0.004	equivalent
Kircher	deploy	8	E0 − disc	11	+0.0031	+0.001 to +0.005	equivalent
Kircher	no_anchor	8	E0 − disc	11	+0.0023	−0.001 to +0.005	equivalent
Kircher	deploy	8	E1 − E0	11	+0.0012	−0.000 to +0.003	equivalent
Kircher	no_anchor	8	E1 − E0	11	−0.0003	−0.003 to +0.002	equivalent
Kircher	deploy	16	E2 − E0	11	−0.0010	−0.004 to +0.002	equivalent
Kircher	no_anchor	16	E2 − E0	11	−0.0001	−0.004 to +0.003	equivalent
Kircher	deploy	16	E2 − disc	11	+0.0029	+0.000 to +0.006	equivalent
Kircher	no_anchor	16	E2 − disc	11	+0.0022	+0.001 to +0.004	equivalent
Kircher	deploy	16	E0 − disc	11	+0.0038	+0.001 to +0.007	equivalent
Kircher	no_anchor	16	E0 − disc	11	+0.0023	−0.001 to +0.006	equivalent
Kircher	deploy	16	E1 − E0	11	+0.0000	−0.003 to +0.004	equivalent
Kircher	no_anchor	16	E1 − E0	11	−0.0013	−0.006 to +0.002	equivalent

Table S6a: EIG convergence check (post hoc), coverage: posterior mass held by the hypotheses kept, median over units (minimum), at the first step and the lowest over the eight steps of a $k = 8$ panel. E0: the 100 most probable hypotheses, 96 quadrature points (as in the main text); E1: 300 hypotheses; E2 (not shown) keeps ≥99% of the mass at every step with 192 points.

Data	View	E0 first	E0 lowest	E1 first	E1 lowest
Siraj	C1	0.98 (0.96)	0.93 (0.79)	0.99 (0.98)	0.98 (0.91)
Siraj	C2	0.98 (0.96)	0.94 (0.76)	0.99 (0.98)	0.98 (0.91)
Siraj	C3	0.61 (0.23)	0.60 (0.23)	0.87 (0.66)	0.86 (0.66)
Kircher	deploy	0.97 (0.97)	0.95 (0.79)	0.99 (0.99)	0.98 (0.92)
Kircher	no_anchor	0.49 (0.32)	0.49 (0.32)	0.84 (0.76)	0.84 (0.76)

Table S7: Label efficiency on Siraj (post hoc, descriptive; 125 loci, 200 repeats). Mean Spearman correlation of the posterior-average prediction after k pilot or random measurements, the label-free default and the best single track chosen with all labels (oracle). Gap closed: share of the default-to-oracle gap recovered by the pilot panel.

View	k	Pilot	Random	Default	Oracle	Gap closed
C1 profiled	2	0.451	0.469	0.474	0.535	-37%
C1 profiled	4	0.469	0.466	0.474	0.535	-8%
C1 profiled	8	0.483	0.467	0.474	0.535	15%
C1 profiled	12	0.489	0.470	0.474	0.535	25%
C1 profiled	16	0.493	0.474	0.474	0.535	31%
C2 related only	2	0.395	0.406	0.412	0.520	-16%
C2 related only	4	0.415	0.403	0.412	0.520	3%
C2 related only	8	0.433	0.406	0.412	0.520	19%
C2 related only	12	0.444	0.413	0.412	0.520	30%
C2 related only	16	0.452	0.420	0.412	0.520	37%
C3 unprofiled	2	0.347	0.180	0.391	0.518	-35%
C3 unprofiled	4	0.402	0.260	0.391	0.518	8%
C3 unprofiled	8	0.438	0.342	0.391	0.518	37%
C3 unprofiled	12	0.453	0.382	0.391	0.518	49%
C3 unprofiled	16	0.461	0.404	0.391	0.518	55%

Table S8: Where the gain is (post hoc, descriptive): Siraj loci split into quartiles of the gap between the label-free default and the oracle track (which needs all labels), gain of the pilot panel over the default at $k = 8$, and loci worse than the default by more than 0.05.

View	Quartile	Loci	Gap	Gain	Share closed	Worse > 0.05
C1 profiled	1	32	0.009	+0.000	0%	0
C1 profiled	2	31	0.030	−0.008	-26%	3
C1 profiled	3	31	0.055	+0.004	8%	0
C1 profiled	4	31	0.155	+0.042	27%	0
C2 related only	1	32	0.057	+0.004	7%	1
C2 related only	2	31	0.085	+0.012	15%	0
C2 related only	3	31	0.108	+0.013	12%	0
C2 related only	4	31	0.184	+0.053	29%	1
C3 unprofiled	1	32	0.053	+0.004	7%	2
C3 unprofiled	2	31	0.090	+0.017	19%	1
C3 unprofiled	3	31	0.121	+0.034	28%	1
C3 unprofiled	4	31	0.244	+0.133	54%	2

Table S9: Cross-locus prior (pre-registered extension; 85 test loci, 20 repeats, $\lambda = 0.75$ chosen on 40 development loci). The prior of a new locus mixes the frozen prior with the mean posterior of m other loci in the same cell type after their own 8-variant pilot panels. X1 (C3, $k = 8$, cross-locus minus single locus at least 0.02): +0.0116, 65/85 loci positive, one-sided Wilcoxon $P = 4 \times 10^{-8}$; not met. X2 (C1, $k = 8$, non-inferior to the matched mean): +0.0141, lower bound +0.0069, 3/85 loci worse by more than 0.05 (upper bound 0.089); met.

View	k	m	Single locus	Cross-locus	Difference	Loci > 0
C3 unprofiled	4	10	0.396	0.418	+0.0217	66/85
C3 unprofiled	8	10	0.436	0.448	+0.0116	65/85
C3 unprofiled	8	1	–	0.442	+0.0061	–
C3 unprofiled	8	3	–	0.446	+0.0095	–
C3 unprofiled	8	30	–	0.449	+0.0127	–
C2 related only	4	10	0.410	0.422	+0.0122	62/85
C2 related only	8	10	0.430	0.443	+0.0121	70/85
C2 related only	8	1	–	0.436	+0.0053	–
C2 related only	8	3	–	0.439	+0.0088	–
C2 related only	8	30	–	0.445	+0.0144	–
C1 profiled	4	10	0.466	0.477	+0.0106	59/85
C1 profiled	8	10	0.484	0.490	+0.0056	62/85
C1 profiled	8	1	–	0.486	+0.0015	–
C1 profiled	8	3	–	0.488	+0.0038	–
C1 profiled	8	30	–	0.491	+0.0068	–

Table S10: Neuron MPRA (pre-registered third dataset; WTC11-NGN2 induced excitatory neurons, 159 regions, 200 repeats). Mean Spearman correlation of TrackPilot after k pilot or random measurements and of the label-free default (matched, related or all-track mean). AlphaGenome: N1 −0.014 (not met), N2 +0.009 ($P = 0.027$, held), N3 −0.019, lower bound −0.028, 47/159 regions worse by more than 0.05 (not met), N4 −0.015 (not met); Enformer: N1 −0.035 (not met), N2 +0.005 ($P = 0.17$, not met), N3 −0.030, lower bound −0.038, 48/159 regions worse by more than 0.05 (not met), N4 −0.021 (not met). Decision (N1 and N3): not supported for either model. The best single track chosen with all labels reached 0.32 and 0.32 (AlphaGenome, Enformer), mostly selection among about 1,000 tracks on about 34 held-out variants per tile.

View	k	AG pilot	AG random	AG default	Enf. pilot	Enf. random	Enf. default
C1 profiled	2	0.052	0.069	0.073	0.042	0.066	0.070
C1 profiled	4	0.054	0.066	0.073	0.040	0.062	0.070
C1 profiled	8	0.053	0.059	0.073	0.040	0.054	0.070
C1 profiled	12	0.054	0.055	0.073	0.041	0.049	0.070
C1 profiled	16	0.053	0.053	0.073	0.041	0.046	0.070
C2 related only	2	0.044	0.057	0.059	0.031	0.050	0.053
C2 related only	4	0.044	0.053	0.059	0.031	0.047	0.053
C2 related only	8	0.044	0.048	0.059	0.032	0.041	0.053
C2 related only	12	0.045	0.045	0.059	0.033	0.038	0.053
C2 related only	16	0.045	0.044	0.059	0.034	0.036	0.053
C3 unprofiled	2	0.024	0.010	0.044	0.010	0.006	0.054
C3 unprofiled	4	0.028	0.013	0.044	0.015	0.009	0.054
C3 unprofiled	8	0.030	0.021	0.044	0.019	0.014	0.054
C3 unprofiled	12	0.032	0.025	0.044	0.021	0.016	0.054
C3 unprofiled	16	0.033	0.027	0.044	0.023	0.018	0.054

Table S11: Applicability check (pre-registered extension; test units, $k = 8$, 20 repeats, threshold $P > 0.5$ chosen on development units). After the pilot panel is measured, a permutation test (200 shuffles of the eight labels) compares the mixture's log evidence with that of shuffled labels; if the labels are explained no better, the label-free default is returned. Mean Spearman correlation of TrackPilot without and with the check, of the default, and the share of runs judged to carry no trackable signal. A1 (neurons: checked − unchecked > 0; checked non-inferior to the default, margin 0.02) and A2 (Siraj: gains over the defaults kept; checked non-inferior to unchecked, margin 0.01) both held; see the repository for the exact statistics. Secondary, neuron test regions with Enformer predictions: checked − unchecked +0.019 ($P = 2 \times 10^{-4}$), checked − default −0.015 (lower bound −0.020).

Data	View	Unchecked	Checked	Default	No signal
Kircher (11 regions)	C1 profiled	0.407	0.410	0.408	5%
Kircher (11 regions)	C3 unprofiled	0.370	0.371	0.302	8%
Neurons (109 regions)	C1 profiled	0.055	0.066	0.072	43%
Neurons (109 regions)	C3 unprofiled	0.026	0.035	0.037	46%
Siraj (85 loci)	C1 profiled	0.484	0.486	0.476	5%
Siraj (85 loci)	C3 unprofiled	0.436	0.438	0.382	8%

Table S12: Threshold of the applicability check (post hoc, descriptive; test units, $k = 8$, 20 repeats). Mean difference in Spearman correlation between the checked prediction and the label-free default for each threshold (fall back to the default when the permutation P exceeds it); "none": no check. The frozen threshold is 0.5; on the development units 0.2, 0.3 and 0.5 scored within 0.0005 of each other. On the neuron assay the permutation P values were close to uniform, as expected when no track carries the signal, so a threshold p^* lets about a share p^* of runs through by chance.

p^*	Siraj C1	Siraj C3	Kircher C1	Kircher C3	Neurons C1	Neurons C3	Neurons C1, Enf.	Neurons C3, Enf.
0.05	+0.003	+0.035	+0.003	+0.031	+0.000	+0.003	−0.000	−0.002
0.1	+0.006	+0.044	+0.005	+0.052	+0.000	+0.003	−0.000	−0.004
0.2	+0.009	+0.052	+0.005	+0.071	−0.000	+0.002	−0.001	−0.007
0.3	+0.010	+0.055	+0.004	+0.073	−0.001	+0.003	−0.003	−0.012
0.5	+0.011	+0.056	+0.002	+0.070	−0.006	−0.002	−0.009	−0.021
none	+0.009	+0.054	−0.001	+0.069	−0.017	−0.011	−0.030	−0.037

Table S13: Kircher test regions measured in human cell lines without tracks of their own (post hoc subgroup; the cell lines were flagged as unprofiled, and a related biosample chosen as their proxy, when the variant table was built, before any run). Frozen TrackPilot, $k = 8$, 200 repeats. With proxy: the proxy tracks are the matched tracks; proxy hidden: proxy and related tracks hidden. AlphaGenome, 5 regions, with proxy tracks: TrackPilot − proxy mean = +0.002 ($3/5 > 0$). AlphaGenome, 5 regions, proxy hidden: TrackPilot − all-track mean = +0.124 ($3/5 > 0$). AlphaGenome, 5 regions, proxy hidden: pilot − random panel = +0.108 ($5/5 > 0$). Enformer, 5 regions, with proxy tracks: TrackPilot − proxy mean = +0.010 ($4/5 > 0$). Enformer, 5 regions, proxy hidden: TrackPilot − all-track mean = +0.160 ($4/5 > 0$). Enformer, 5 regions, proxy hidden: pilot − random panel = +0.101 ($5/5 > 0$).

Model	Region	Cell line	Proxy mean	TP, with proxy	All-track mean	TP, proxy hidden	Random panel
AlphaGenome	GP1BA	HEL 92.1.7	0.498	0.498	0.374	0.481	0.329
AlphaGenome	HBB	HEL 92.1.7	0.645	0.645	0.480	0.450	0.378
AlphaGenome	HBG1	HEL 92.1.7	0.519	0.529	0.182	0.545	0.399
AlphaGenome	IRF4	SK-MEL-28	0.650	0.650	0.340	0.562	0.434
AlphaGenome	TERT (glioblastoma)	glioblastoma (alt.)	0.641	0.642	0.666	0.623	0.584
Enformer	GP1BA	HEL 92.1.7	0.387	0.388	0.289	0.402	0.205
Enformer	HBB	HEL 92.1.7	0.492	0.496	0.383	0.474	0.364
Enformer	HBG1	HEL 92.1.7	0.292	0.337	−0.119	0.343	0.290
Enformer	IRF4	SK-MEL-28	0.576	0.576	0.298	0.472	0.358
Enformer	TERT (glioblastoma)	glioblastoma (alt.)	0.594	0.592	0.616	0.576	0.546

Table S14: Pre-registered test: Kircher experiments measured in mouse cell lines, which no human track profiles (five regions; the two ZRS experiments averaged). A human proxy was named from metadata before any scoring (neuroblastoma lines for Neuro-2a, endocrine-pancreas precursors for MIN6, fibroblasts for NIH/3T3). Frozen TrackPilot, $k = 8$, 200 repeats. Hypotheses: M1, proxy hidden, TP − all-track mean ≥ 0.03 and $\geq 4/5$ regions > 0; M2, proxy hidden, pilot − random panel $\geq 4/5$ regions > 0; M3, with proxy, TP − proxy mean ≥ -0.02 and no region below -0.05; supported if M1 and M3 hold. AlphaGenome: M1 fails, M2 holds, M3 fails; Enformer: M1 fails, M2 holds, M3 holds. Neither model met the pre-registered criterion. Best track: best single track chosen with all labels (post hoc).

Model	Region	Cell line	Proxy mean	TP, with proxy	All-track mean	TP, proxy hidden	Random panel	Best track
AlphaGenome	RET	Neuro-2a	0.237	0.170	0.106	0.060	0.071	0.256
AlphaGenome	TCF7L2	MIN6	0.136	0.143	0.139	0.168	0.085	0.255
AlphaGenome	UC88	Neuro-2a	0.327	0.332	0.334	0.318	0.189	0.356
AlphaGenome	ZFAND3	MIN6	0.240	0.406	0.426	0.451	0.341	0.513
AlphaGenome	ZRS	NIH/3T3	0.223	0.235	0.228	0.216	0.097	0.305
Enformer	RET	Neuro-2a	0.228	0.184	0.172	0.165	0.086	0.256
Enformer	TCF7L2	MIN6	0.115	0.115	0.105	0.117	0.041	0.204
Enformer	UC88	Neuro-2a	0.213	0.213	0.185	0.202	0.095	0.283
Enformer	ZFAND3	MIN6	0.186	0.321	0.302	0.423	0.273	0.483
Enformer	ZRS	NIH/3T3	0.219	0.214	0.203	0.169	0.068	0.281

Table S15: Frozen TrackPilot on two models. Siraj: 40 random loci, both cell types averaged; Kircher: 11 test regions, unprofiled scenario. Differences in Spearman correlation on held-out variants; regions or loci > 0 and one-sided P in brackets; [a]unprofiled, [b]related cells only. Kircher random-panel P: 0.002 (AlphaGenome), 0.001 (Enformer). Both columns use the frozen method with the family prior; Table 1 of the main text shows the pre-registered version. Numbers copied from the decision files named in the source.

TrackPilot minus	AlphaGenome	Enformer
Siraj, 40 loci		
matched mean, $k = 8$	+0.006	+0.022
loci worse by > 0.05	1/40	1/40
all-track mean[a], $k = 8$	+0.032 (3e−4)	+0.052 (4e−4)
related mean[b], $k = 16$	+0.036 (4e−10)	+0.047 (5e−7)
matched mean (absolute)	0.473	0.404
Kircher, 11 regions[a], $k = 8$		
random panel	+0.090 (9/11)	+0.097 (10/11)
all-track mean	+0.077 (8/11)	+0.109 (8/11)
TrackPilot (absolute)	0.373	0.324